

Section 8

① Each mode $\mathbf{k}$ of φ is an oscillator of unit mass with $\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$. So m is now the particle mass, not the oscillator mass set to 1 in LecNote6, and ω depends on $\mathbf{k}$ (p.3).

The Path Integral for free field theory.

Our result for the one-dimensional simple harmonic oscillator can be transported into a free field theory with hamiltonian density

$$\mathcal{H}_0 = \frac{1}{2} \Pi^2 + \frac{1}{2} (\nabla\varphi)^2 + \frac{1}{2} m^2 \varphi^2$$

$g(t) \rightarrow \varphi(\vec{x}, t)$ (classical field)

$Q(t) \rightarrow \varphi(\vec{x}, t)$ (operator field)

$f(t) \rightarrow J(\vec{x}, t)$ (classical source)

① their distinction resides in the context.

using the 1- $i\epsilon$ trick, we need to shift ②

$$m^2 \rightarrow (1-i\epsilon)m^2$$

From now on every m^2 means we multiplied by 1- $i\epsilon$ already

Now we can write down the path integral:

(also called the functional integral of our free field theory)

$$Z_0(J) \equiv \langle 0|0\rangle_J = \int \mathcal{D}\varphi e^{i \int d^4x [\mathcal{L}_0 + J\varphi]}$$

with

$$\mathcal{L}_0 = -\frac{1}{2} \partial^\mu \varphi \partial_\mu \varphi - \frac{1}{2} m^2 \varphi^2$$

② $(1-i\epsilon)H_0$ damps all three terms; after the Π integral the kernel is $-(k^2 + m^2) + i\epsilon(k_0^2 + \mathbf{k}^2 + m^2)$, i.e. $m^2 \rightarrow m^2 - i\epsilon$ (cf. LecNote6 p.2). Writing $(1-i\epsilon)m^2$ is equivalent for $m \neq 0$ but gives nothing at $m=0$.

and $\mathcal{D}\varphi \propto \prod_x d\varphi(x)$ the Functional Measure

Now every "Path" in the path integral mean a path in the space of field configuration.

We can evaluate $Z_0(J)$ by mimicking SFTO.

$$\tilde{\varphi}(k) = \int d^4x e^{-ikx} \varphi(x)$$

$$\varphi(x) = \int \frac{d^4k}{(2\pi)^4} e^{ikx} \tilde{\varphi}(k)$$

2 As in LecNote6 p. 2, the χ integral is J -independent, and $Z_0(0) = 1$ normalizes $\mathcal{D}\varphi$. The $-i\epsilon$ in m^2 (p. 1) makes it converge:
 $e^{iS_0} \supset e^{-\frac{\epsilon}{2} \int \frac{d^4k}{(2\pi)^4} |\tilde{\chi}|^2}$

Starting with $S_0 = \int d^4x [L_0 + J\varphi]$, we get

$$S_0 = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} [-\tilde{\varphi}(k)(k^2 + m^2)\tilde{\varphi}(-k) + \tilde{J}(k)\tilde{\varphi}(-k) + \tilde{J}(-k)\tilde{\varphi}(k)]$$

Changing the path integration variable

$$\tilde{\chi}(k) = \tilde{\varphi}(k) - \frac{\tilde{J}(k)}{k^2 + m^2} \quad (1)$$

Since $\frac{-\tilde{J}(k)}{k^2 + m^2}$ is a constant in the functional space

$$\mathcal{D}\varphi = \mathcal{D}\chi$$

1 Same shift as LecNote6 p. 2, despite the sign: the kernel here is $-(k^2 + m^2) = k_0^2 - \omega_k^2$, the analogue of $E^2 - \omega^2$ there, so $-\tilde{J}/(k^2 + m^2) = +\tilde{J}/(k_0^2 - \omega_k^2)$.

$$S_0 = \frac{1}{2} \int \frac{d^4k}{(2\pi)^4} \left[\frac{\tilde{J}(k)\tilde{J}(-k)}{k^2 + m^2} - \tilde{\chi}(k)(k^2 + m^2)\tilde{\chi}(-k) \right]$$

Similarly, $Z_0(0) = \langle 0|0 \rangle_{J=0} = 1. \quad (2)$

Hence

$$Z_0(J) = e^{\frac{i}{2} \int \frac{d^4k}{(2\pi)^4} \frac{\tilde{J}(k)\tilde{J}(-k)}{k^2 + m^2 - i\epsilon}}$$

$$= e^{\frac{i}{2} \int d^4x d^4x' J(x) \Delta(x-x') J(x')}$$

③

Here we have the Feynman propagator

$$\Delta(x-x') = \int \frac{d^4 k}{(2\pi)^4} \frac{e^{ik(x-x')}}{k^2 + m^2 - i\epsilon}$$

which is ~~the~~ a Green's function for the Klein-Gordon equation

$$(-\partial_x^2 + m^2) \Delta(x-x') = \delta^4(x-x')$$

1 $k^2 + m^2 - i\epsilon = -(k^0)^2 + \omega^2 - i\epsilon$ with $\omega = \sqrt{k^2 + m^2}$, so the k^0 integral is G of LecNote6 p. 3, $\frac{i}{2\omega} e^{-i\omega|t-t'|}$. Its $\frac{1}{2\omega}$ joins $\frac{d^3 k}{(2\pi)^3}$ into $\widetilde{dk}$.

Again performing a contour integral, we get.

$$\begin{aligned} \Delta(x-x') &= i \int \widetilde{dk} e^{i\vec{k}(\vec{x}-\vec{x}') - i\omega|t-t'|} \leftarrow \text{similar to previous example, the contour choice changes with time order} \\ &= i\theta(t-t') \int \widetilde{dk} e^{ik(x-x')} + i\theta(t'-t) \int \widetilde{dk} e^{-ik(x-x')} \end{aligned}$$

$\theta(t)$ is the unit step function.

2 Here k is on shell, $k^0 = \omega$: $e^{ik(x-x')} = e^{i\vec{k}\cdot(\vec{x}-\vec{x}') - i\omega(t-t')}$, not the off-shell $d^4 k$ variable above. The $t < t'$ term uses $\vec{k} \rightarrow -\vec{k}$, allowed since dk is invariant.

Now we should have

$$\langle 0 | T \phi(x_1) \dots | 0 \rangle = \frac{1}{i} \frac{\delta}{\delta J(x_1)} \dots Z_0[J] \Big|_{J=0}$$

with explicit $Z_0[J]$ expression, we have

$$\langle 0 | T \phi(x_1) \phi(x_2) | 0 \rangle = \frac{1}{i} \frac{\delta}{\delta J(x_1)} \frac{1}{i} \frac{\delta}{\delta J(x_2)} Z_0[J] \Big|_{J=0}$$

$$= \frac{1}{i} \frac{\delta}{\delta J(x_1)} \left[\int d^4 x' \Delta(x_2 - x') J(x') \right] Z_0[J] \Big|_{J=0}$$

$$= \frac{1}{i} \frac{\delta}{\delta J(x_1)} \left[\frac{1}{i} \Delta(x_2 - x_1) + \int \dots J(x') \dots J(x'') \right] Z_0[J] \Big|_{J=0}$$

$$= \frac{1}{i} \Delta(x_2 - x_1)$$

Consequently

$$\begin{aligned} & \langle 0 | T \varphi(x_1) \varphi(x_2) \varphi(x_3) \varphi(x_4) | 0 \rangle \\ &= \frac{1}{i^4} [\Delta(x_1 - x_2) \Delta(x_3 - x_4) + \Delta(x_1 - x_3) \Delta(x_2 - x_4) \\ & \quad + \Delta(x_1 - x_4) \Delta(x_2 - x_3)]. \end{aligned}$$

and generally

$$\begin{aligned} & \langle 0 | T \varphi(x_1) \dots \varphi(x_{2n}) | 0 \rangle \\ &= \frac{1}{i^n} \sum_{\substack{\{i, j\} \\ \text{pairings}}} \Delta(x_{i_1} - x_{j_1}) \dots \Delta(x_{i_n} - x_{j_n}). \end{aligned}$$

Also known as Wick's theorem.